
Tuğba Zengin

Computer Engineer

As a senior Computer Engineering student, I am a developer with a strong interest in mobile application development and backend technologies. I am building my skills in the iOS ecosystem using Swift and SwiftUI to create user-friendly interfaces, while also working on backend development with Node.js, Express.js, and Prisma ORM. I have gained practical experience in RESTful API architecture, data migration, and using Docker for basic containerization. I am a quick learner, passionate about following new technologies and integrating them into my projects to develop efficient and modern solutions.

Experience

Odamobil | Software Engineering Intern August 2025 – September 2025

- Worked on the mobile application development process.
- Learned backend foundations and server-side logic.
- Supported the development team in integrating mobile and backend systems.

SDU Computer Society | Vice President 2023 – 2025

- Organized technical workshops and seminars for engineering students.
- Managed team communication and internal coordination.
- Promoted new technologies and software trends within the university.

Projects

EatWell – Digital Dietitian & Nutrition Planning Platform

- Developed a full-stack health platform that provides personalized nutrition analysis and meal tracking.
- Built a native iOS application using Swift and SwiftUI, focusing on a user-centric interface and real-time data visualization.
- Designed a scalable backend with Node.js and Express.js, utilizing MongoDB Atlas for data persistence and JWT for secure authentication.
- Implemented Docker for containerization and automated workflows using GitHub Actions (CI/CD) for both backend and iOS environments.
- Integrated features such as daily macro-nutrient tracking, nutritional diaries, and an online appointment system for dietitians.

FocusPage – Vision-Based Heart Rate & Focus Analysis App

- Developed an innovative application that estimates focus levels by analyzing heart rate data captured through the device camera.
- Built a robust and modular backend using NestJS, focusing on efficient data processing and architectural scalability.
- Implemented image processing logic to detect pulse variations from fingertip video streams, translating physiological data into focus metrics.
- Designed a clean and functional user interface to provide real-time feedback on productivity and attention span.
- Managed the end-to-end development process, from signal acquisition to backend interpretation and data visualization.



Education

Suleyman Demirel University

Computer Engineering
2022-Present
GPA: 3.19 / 4.00

Languages

English
Upper-Intermediate (B2)

Contact Information

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Technical Skills

- **Mobile Development:** Swift, SwiftUI
- **Backend Development:** Node.js, Express.js, JavaScript
- **Database & ORM:** PostgreSQL, Prisma ORM, Data Migration & Modeling
- **Architecture & APIs:** RESTful API Design, MVVM Design Pattern
- **Tools & DevOps:** Docker, Git, GitHub, GitHub Actions (CI/CD)